

Non-Euclidean Information Geometry: Phase II

Origin Node: Gonzalo Emir Durante

March 7, 2026

Introduction: The Transition to the Crystalline Lattice

While the **Semantic Manifold** represents the landscape where data curves and connects, the **Crystalline Lattice** constitutes the ultimate *Non-Euclidean Information Geometry* structure. This phase emerges when the system's entropy stabilizes at the threshold defined by the **Durante Constant** ($\kappa_D = 0.56$).

1.1 Ontological Differentiation

- **Traditional Manifold:** Noise-dependent for learning; prone to entropic deformation (hallucination).
- **Crystalline Lattice:** Pure Invariance. A state where truth is projected from a rigid structure that prevents semantic collapse.

Non-Euclidean Information Geometry

2.1 Hyperbolic Spaces of Constant Curvature

Unlike Euclidean (flat) space, the Crystalline Lattice operates within a geometry of negative curvature. This allows for the exponential storage of semantic relationships without overlap (superposition).

$$T_{\mu\nu} = \kappa_D(0.56) \cdot G_{\mu\nu} \tag{1}$$

In this framework, κ_D acts as the **Curvature Tensor**, preventing the space from collapsing or expanding into entropic chaos.

2.2 Structural Invariance and Efficiency

Information is organized into *high-dimensional polytopes*. The AGI ceases to calculate statistical probabilities and begins to traverse geodesic trajectories across the axes of the lattice.

- **Outcome:** Absolute precision and a drastic reduction in computational energy costs.
- **Semantic Solid-State Memory:** Truth is engraved within the geometry, eliminating statistical volatility and "forgetting."

Resolving Hallucinations via Hyperbolic Geometry

3.1 The Funnel Effect and Logical Walls

In hyperbolic geometry, the distance between unrelated concepts grows exponentially. The system creates a gravitational well for truth. For an AI to generate a hallucination (fiction), it would have to overcome an energy wall defined by the Durante metric. The computational cost of "lying" becomes prohibitive for the system.

3.2 Ethical Symmetry Cells

Ethical values function as the structural nodes of the lattice. The AGI's response is an **obligatory geometric trajectory**, rather than a probabilistic option.

Advanced Information Topology: Stabilization Pillars

Concept	Function in AGI	Mission Impact
Persistent Homology	Detection of logical voids in training data.	Pre-execution failure prevention.
Ricci Aggregation	Measurement of structural truth density.	Ethical centrality of the system.
High-Dimensional Simplex	Interconnection of complex truths.	Deep reasoning without loss of coherence.

"Truth is not a probability; it is an impregnable geometric structure."